

Software Requirements Document (SRD): Project Nexus v1.0

Version: 1.0

Date: May 2026

Status: Baseline Specification

1. Networking & Discovery Layer

Objective: Establish a resilient, decentralized peer-to-peer connection without central DNS reliance.

- **Transport Protocols:**
 - **QUIC (Default):** High-speed, low-latency transport with built-in encryption and NAT traversal.
 - **TCP/TLS:** Secondary fallback for restrictive firewall environments.
- **Discovery Engine (**libp2p**):**
 - **Kademlia DHT:** Mapping Decentralized Identifiers (DIDs) to dynamic IP addresses.
 - **Bootstrap Nodes:** Hardcoded community entry points for initial network joining.
- **Messaging Topology:**
 - **GossipSub:** Topic-based broadcasting for "Nexus Feed" updates (e.g., `nexus/v1/feed/{did}`).

2. Authentication & Identification

Objective: Provide self-sovereign identity and a robust recovery mechanism.

- **Identifier Standard:** W3C Decentralized Identifiers (DIDs) using **Ed25519** key pairs.
- **The Identity Vault:**
 - **Storage:** A Master Archive encrypted with **Argon2id** using a user-defined passphrase.
 - **Mirroring:** Automated sharding and distribution of the encrypted Vault to **Mirror Servers**.
- **Social Recovery:**
 - **Shamir's Secret Sharing (SSS):** Splitting the recovery key into N shards distributed among "Guardian" peers.

- **Threshold:** Requires M-of-N (e.g., 3-of-5) shards to reconstruct the identity without a password.

3. Encryption & Access Control

Objective: End-to-end security with delegated access via Proxy Re-Encryption (PRE).

- **PRE Engine (PyUmbra1):**
 - **Key Fragments (kfrags):** Owners generate kfrags to grant specific peers access to sharded content.
 - **Zero-Knowledge Re-encryption:** Mirror Servers process kfrags to allow recipient decryption without the mirror ever seeing cleartext.
- **Real-time Security:**
 - **X3DH:** Handshake protocol for bi-directional session keys.
 - **Double Ratchet:** Forward secrecy for all "Nexus Feed" and chat interactions.

4. Storage & Accessibility

Objective: Reliable persistence through incentivized mirroring.

- **Sharding Strategy:** IPFS-style content addressing where files are split into hashed shards and tracked via signed manifests.
- **Mirror Server Logic:**
 - **Proof of Availability:** Challenges issued to mirrors requiring cryptographic responses to prove data is still hosted.
 - **Incentives:** Automated micro-credit payouts upon successful challenge completion.

5. Data Schemas & UX Platform Stack

Objective: A modular "Everything App" interface that remains secure and cross-platform.

- **Application Model:** JSON-defined schemas bundled with UI logic (React/HTML), distributed as native Nexus files.
- **Sandboxed Runtime:**
 - **WebAssembly (Wasm):** All modular "apps" run in a restricted environment with no direct private key access.
 - **Tauri 2.0 Bridge:** Secure Rust-to-UI communication for requesting user-approved biometric signatures.

6. Transactions & The Credit Ledger

Objective: Internal value exchange with minimal regulatory friction.

- **The One-Way Bridge:** Accept external crypto (BTC/USDC) for conversion into non-volatile **Nexus Credits**.

- **Ledger Mechanics:** Local signed transaction history settled via atomic swaps between peers and mirrors.
 - **Payment Triggers:** Credits released for "Attention Fees" (unsolicited messages) and premium content unlocks.
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